AFS Fiberwise

First technical assessment for in-line spectroscopy

AFS evidence for a reviewable first experiment

TEAM

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github.com/ght007/SchubKI
https://schubki-frontend-livid.vercel.app/

The first customer discussion

The assistant proposes an investigation for an AFS expert to review.

01 **Customer enquiry**

Target, material, process conditions and existing equipment.

02 **Confirm the reading**

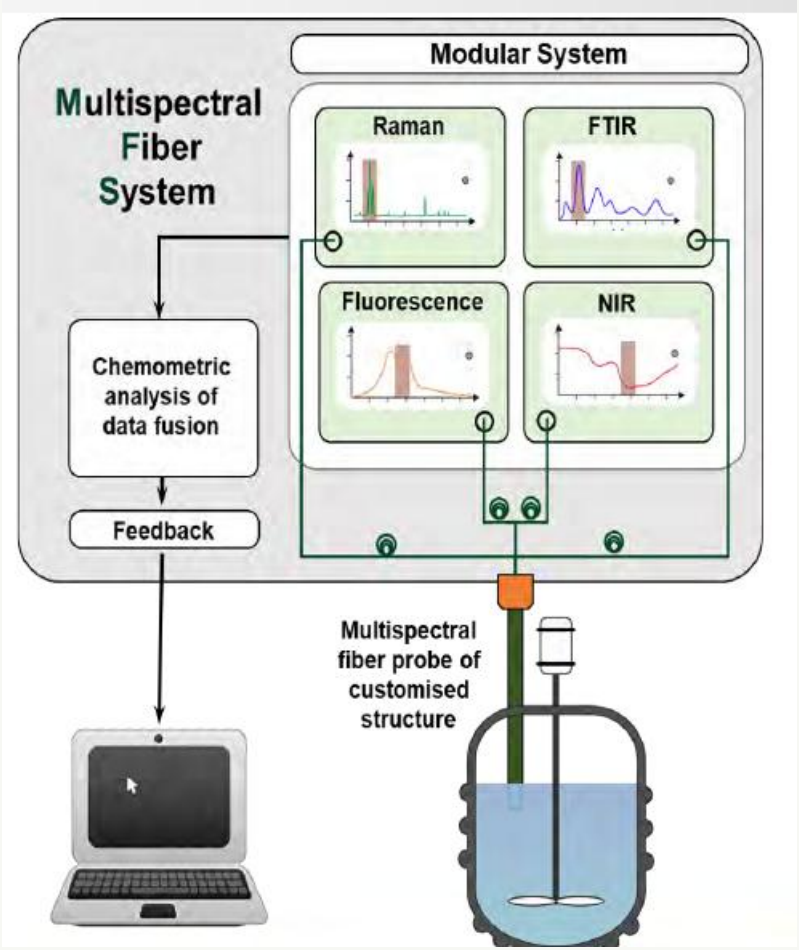
Confirm extracted facts and correct misreadings.

03 **Technical assessment**

Review methods, evidence and a first experiment.

04 **Expert follow-up**

Add customer answers and compare the revised assessment.

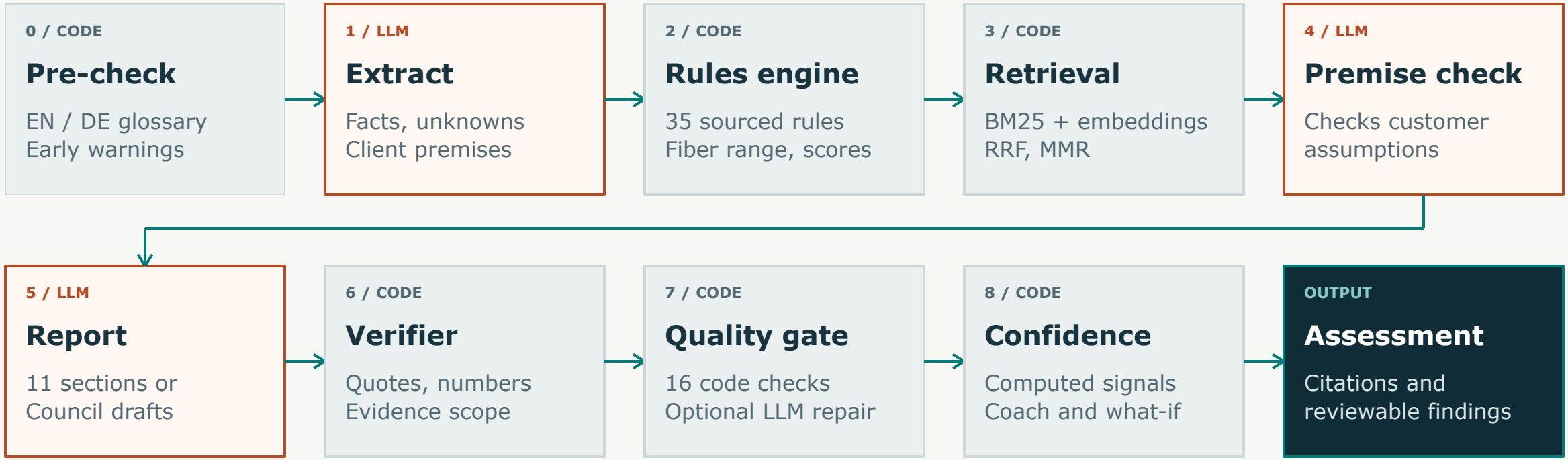


AFS multispectral laboratory concept
Source: S02, slide 5

Sources: challenge brief p.1; S02 slide 20. Workflow: frontend/js/app.js.

System architecture

Rules handle the physics. Models handle language. Code checks the drafted report.



Knowledge base: 95 verbatim evidence cards, 35 sourced rules, method profiles, fibers and bands

Orange border: model stage. Code runs the checks. Optional repair and Coach also use a model.

Adapted from AFS Fiberwise - Video Intro, slide 2. Implementation: engine/pipeline.py, precheck.py and confidence.py.

AFS materials become inspectable evidence

Page references and evidence scope stay attached to the knowledge.

95
curated evidence cards

35
general sourced rules

16
quality checks

S01 + comparison table	Methods, interfaces, calibration and application examples.
S02 process-control slides	Fiber transmission, analyte bands, sensor concepts and A/B/C pathways.
S03 + S04 research papers	Laboratory evidence with explicit limits on readiness claims.
S05 + S06 business context	Company and laboratory context only.

S02 uses a labelled slide transcription. Ingest checks evidence-card quotes.

Sources: data/kb/claims.jsonl, rules.yaml and sources.json.

Verification supports expert review

Statement types and checks expose uncertainty.

Four statement types

Client fact — What the customer actually said.

Sourced claim — A claim with a source, page and quote.

Assumption — An unconfirmed interpretation.

Proposal — An experiment or design to investigate.

Checks after drafting

Citation and scope

Quote presence, missing pages and readiness language.

Numbers and client facts

Heuristic checks against supplied text and evidence.

Quality and repair

16 checks, with at most one model repair attempt.

Confidence uses information, evidence and scope signals, with rule caps and verification penalties.

A matching quote does not prove that the full scientific conclusion is correct.

Code: engine/verify.py, quality.py and confidence.py. Citation counts cover selected report fields.

Example: a new pharmaceutical formulation

H01 asks whether adding Raman can preserve an existing NIR calibration after the formulation changes.

Customer premise (paraphrased)

“Adding Raman will keep our old calibration valid.”

Assessment finding

Adding Raman does not validate the old model for a new formulation.

Methods worth testing	NIR diffuse reflection as the baseline, with Raman as a comparison.
Decision-changing questions	What changed in the blend? Which reference values and acceptance criteria are available?

Archived H01: provisional assessment, low confidence. The report stage used a rule-based fallback.

A first feasibility experiment

The first experiment tests whether calibration reuse is justified.

01 **Representative samples**

Cover the new API / excipient range and relevant physical variation.

02 **Independent reference values**

Pair each sample spectrum with a suitable laboratory measurement.

03 **Controlled comparison**

Test the old NIR model first. Compare a new NIR model with Raman or fused data if justified.

04 **Independent validation**

Evaluate held-out batches or runs against client-agreed error and bias.

Decision: retain, rebuild or reject calibration reuse using the experimental evidence.

Based on the stored H01 plan. Acceptance criteria remain client-agreed; no spectra were supplied.

Example: the existing fiber blocks the MIR concept

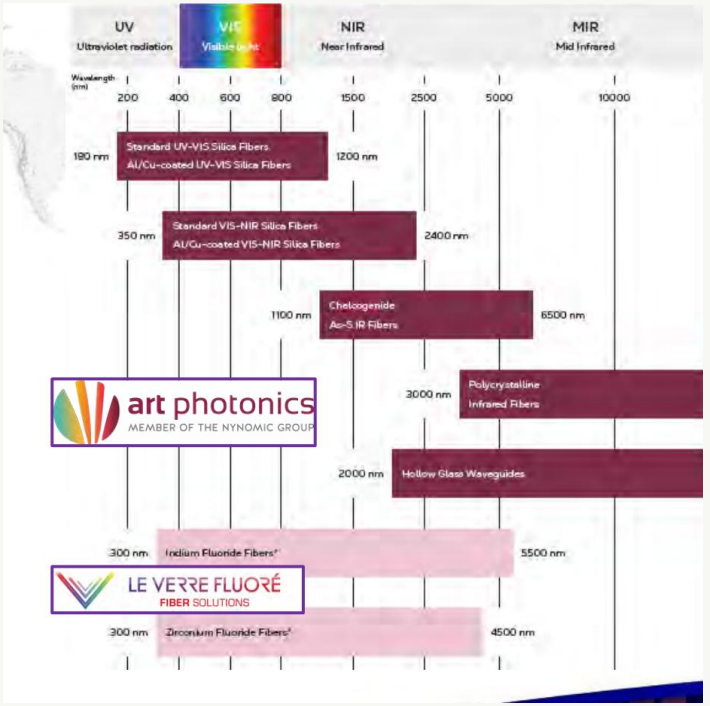
H04 asks to reuse a visible/NIR silica cable for remote MIR ATR measurements of water in petroleum.

The central finding

The existing silica link cannot support the proposed MIR measurement.

Next questions: link length, routing, process conditions, target range and reference measurements.

Next experiment: compare a compatible MIR setup on representative samples before approving a remote link.



S02 slide 6, cropped fiber transmission chart

Archived H04 also flags 2/7 sourced claims verified and one unsupported number. Expert review is still required.

Archived H04 output. Fiber chart: art photonics, supplied S02 slide 6.

AFS pathways follow the investigation

Optional pathway preferences are checked against sourced rules.

Lab trials establish useful features before solution selection.

A

Chemometric model

Translate spectra into process measurements for automation.

B

Customized MSF system

Equip the customer’s R&D laboratory.

C

Dedicated spectral sensors

Develop a dedicated sensor after lab validation.

A preference is context. The analysis can return “premature”, “conflicts” or “undecided”.

Path definitions: S02 slide 20. Code: engine/paths.py. AFS and the customer confirm the solution.

Current limits and the expert’s role

The prototype structures the first decision. Real samples determine analytical performance.

No measured spectra	Accuracy and detection limits remain unproven for the customer’s process.
Heuristic safeguards	Rule scores and confidence are design signals, not calibrated probabilities.
Incomplete verification	Quote checks do not prove conclusions. Citation counts cover selected fields.
Model and service failures	Retries and labelled fallbacks help when model calls fail.

Expert review and feasibility studies guide the next step.

Saved-run evidence and evaluation scope

These are reported artifact metrics, not independently measured scientific accuracy.

Archived run	Quality checks	Sourced claims	Unsupported numbers
H01 formulation	**14 / 16**	**10 / 10**	**0**
H04 silica / MIR	**14 / 16**	**2 / 7**	**1**

What the artifacts demonstrate

H01 retains a labelled rule-based draft after a model failure. H04 exposes citation and numeric failures rather than presenting a clean verification result.

What they do not establish

The reviewed snapshot contains two final-case outputs and six adversarial outputs. It does not establish completion of H01-H06 or expert-validated performance.

Evidence: outputs/H01.json, H04.json and outputs/adversarial at snapshot bf818a9. H01: H1 fail / U2 warn. H04: G1 and G2 fail.

Sources and reproducibility

Public source code and traceable materials support jury review.

Implementation	github.com/ght007/SchubKI Snapshot bf818a9, inspected on 18 October 2026.
Core files	engine/pipeline.py, paths.py, verify.py and confidence.py kb/retrieve.py, kb/store.py and data/kb/*
AFS evidence	S01: methods and calibration, pp.5-10 and 14-18 S02: fiber chart, slide 6; lab concept, slide 5; pathways, slide 20
Evidence scope	S03: LED kidney study, in-vitro proof of concept S04: combined MIR / fluorescence probe, laboratory demonstration

Reproduction: Python 3.12, requirements.txt, registered AFS source files, ingest and local app startup. Groq requires a private API key.

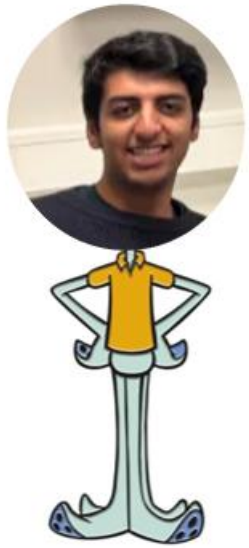
The repository README documents setup. The original AFS documents retain their owners' rights.

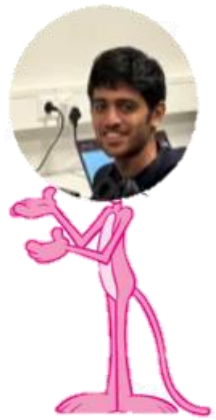
Challenge context: AI Assistant for In-line Spectroscopy Solution Design brief, pp.1-2; opening presentation, p.22.

TEAM











-Shades-